Exercise 1 : SQL Fundamentals

Question 1 : Select statement

Retrieve all columns from the employee table.

```
SELECT *
FROM employees ;
```

id	first_name	last_name	department	Salary	hire_date	city
1	John	Doe	IT	55 000	2018-06-15	New york
2	Jane	Smith	HR	48 000	2019-07-20	Chicago
3	Mike	Johnson	Finance	60 000	2017-09-20	Los Angeles
4	Sarah	Brown	IT	53 000	2021-03-25	New york
5	David	White	Marketing	52 000	2016-04-10	San fransisco
6	Emily	Davis	IT	62 000	2015-02-14	Chicago
7	Robert	Wilson	finance	59 000	2019-01-01	Houston
8	Jessica	Moore	HR	51 000	2018-03-22	Los Angeles
9	Daniel	Clark	Marketing	53 000	2022-06-01	Chicago
10	Laura	Hall	IT	50 000	2020-08-10	San francisco

Question 2 : Select distinct

find all unique departments in the employees table.

```
SELECT DISTINCT department
FROM employees ;
```

department
IT
HR
Finance
Marketing

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Question 3: ORDER BY Statement

Retrieve all employees first & last names, ~~ordered~~ ordered by salary in descending order.

```
SELECT first_name,
       last_name,
       salary
FROM employees
ORDER BY salary DESC;
```

first_name	last_name	Salary
Emily	Davis	62 000
Mike	Johnson	60 000
Robert	Wilson	59 000

Question 4 LIMIT Statement

Retrieve the top 5 highest-paid employees.

```
SELECT first_name,
       last_name,
       salary
FROM employees
ORDER BY salary DESC
LIMIT 5;
```

First_name	last_name	Salary
Emily	Davis	62000
Mike	Johnson	60 000
Robert	Wilson	59 000
John	Doe	55000
Sarah	Brown	53 000

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Question 5 : WHERE Statement.

Find employees who work in the IT department

```
SELECT first-name,
       last-name,
       department
FROM employees
WHERE department = 'IT';
```

first-name	last-name	department
John	Doe	IT
Sarah	Brown	IT
Emily	Davis	IT
Laura	Hall	IT

Question 6 : AND Statement.

find employees who work in finance and have salary greater than 58,000.

```
SELECT first-name,
       last-name,
       department,
       Salary
FROM employees
WHERE department = 'Finance' AND salary > 58 000;
```

first-name	last-name	department	salary
Mike	Johnson	Finance	60000
Robert	Wilson	Finance	59000

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Question 7 : OR Statement

find employees who work in the HR department OR Marketing dep

```
SELECT first_name,
       last_name,
       department
FROM employees
WHERE department = 'HR' OR department = 'Marketing';
```

first_name	last_name	department
Jane	Smith	HR
David	White	Marketing
Jessica	Moore	HR
Daniel	Clark	Marketing

Question 8 NOT Statement

find employees who do not work in the IT department.

```
SELECT first_name,
       last_name,
       department
FROM employees
WHERE NOT department = 'IT';
```

first_name	last_name	department
Jane	Smith	HR
Mike	Johnson	finance
David	White	Marketing
Robert	Wilson	finance
Jessica	Moore	HR
Daniel	Clark	Marketing

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Question 9 : IN statement

find employees who are in the HR, IT or Finance dept.

```
SELECT first_name,
       last_name,
       department
FROM employees
~~WHERE~~
WHERE department IN ('HR', 'IT', 'Finance');
```

first_name	last_name	department
John	Doe	IT
Jane	Smith	HR
Robert	Wilson	Finance
Mike	Johnson	finance
Sarah	Brown	IT
Emily	Davis	IT
Jessica	Moore	HR
Laura	Hall	IT

Question 10 : combining conditions

find employees who are in the IT department, have salary > 50000 & located in New-York. ~~City~~.

```
SELECT first_name,
       last_name,
       department,
       Salary,
       city
From employees
WHERE department = 'IT'
AND salary > 50 000
AND city = 'New York';
```

first_name	last_name	department	Salary	City
John	Doe	IT	55000	New York
Sarah	Brown	IT	53000	New York

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Question 11 : combining WHERE, AND, and ORDER BY.

Retrieve all employees who work in the finance or marketing department, earn more than 52 000 and order the results by salary in ~~des~~ descending order.

```
SELECT first_name,
       last_name,
       department,
       salary
FROM employees
WHERE department = 'Finance' OR department = 'Marketing')
AND salary > 52 000
ORDER BY salary DESC;
```

first_name	last_name	department	Salary
Mike	Johnson	finance	60000
Robert	Wilson	Finance	59000
Daniel	Clark	Marketing	53000

Question 12 : Combining SELECT DISTINCT, WHERE and IN.

```
SELECT DISTINCT city, first_name, last_name, departments
FROM employees
WHERE department NOT IN ('IT', 'HR');
```

first_name	last_name	department	City
Mike	Johnson	Finance	Los Angeles
David	White	Marketing	San Fransisco
Robert	Wilson	Finance	Houston
Daniel	Clark	Marketing	Chicago

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Question 13: Combining WHERE, NOT, AND, and Order By

Retrieve employees who are NOT in the finance department, have salary > 50 000 & order results by hire date in ascending order.

```
SELECT first_name,
       last_name,
       department,
       Salary,
       hire_date
FROM employees
WHERE department != 'Finance'
AND Salary > 50 000
ORDER BY hire_date ASC;
```

first_name	last_name	department	Salary	hire_date
Emily	Davis	IT	62 000	2015-02-14
David	White	Marketing	57 000	2016-04-10
Jessica	Moore	HR	51 000	2018-05-22
John	Doe	IT	55000	2018-06-15
Sarah	Brown	IT	53 000	2021-03-25

Question 14 : Combining WHERE, OR, IN, and LIMIT

Retrieve first 3 employees who work in either Chicago or Los Angeles & belong to the IT or Marketing department.

```
SELECT first_name,
       last_name,
       department,
       city
FROM employees
WHERE city IN ('Chicago', 'Los Angeles')
AND department IN ('IT', 'Marketing')
LIMIT 3;
```

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first_name	last_name	department	city
Jane	Smith	HR	Chicago
Mike	Johnson	Finance	Los Angeles
Emily	Davis	IT	Chicago

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